

Operations Research

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1 Wednesday April 22nd

1.1 Exercises

1.1.1 Graph Question 4

Show that a connected graph has at least $n - 1$ edges.

In the best case scenario, if we build a graph starting from one node, in order to connect it we would like to connect it and reduce the number of connected components by one each time. Therefore, $C(G_n) = C(G_{n-1}) - 1 = n - 1$. A tree, which has no cycles and is connected, must have exactly $n - 1$ edges.

1.1.2 Algorithm 1

Prove: it is always possible to extract, from a (directed) path going from i to j , a simple path going from i to j .

A simple path is defined as a path where no node is visited more than once. Effectively, from any path, that can contain cycles, it is possible to create an acyclic path. For any cycle, we can have the same path but exit the cycle at the node previous to the cycling node. For a cycle and a path entering that cycle, we can define the nodes N_i, N_o, N_c as the input, output, and cycling node. We can always take the node that feeds into the cycline node $N_c - 1$, and use that as the node to feed into the output.

1.1.3 Algorithm 2

Determining the adjacency matrix, alpha, and beta for the graph

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \quad (1)$$

$$\beta = [1, 2, 3, 4, 1, 3, 5, 3, 4] \quad (2)$$

$$\alpha = [1, 5, 7, 7, 8, 10] \quad (3)$$

To extract the list of successors of a node n , we get the list $[\beta[\alpha[n]], \beta[\alpha[n + 1] - 1]]$

1.1.4 Bipartite

For a bipartite and bicoloured graph, any green node can be reached from a starting red node in a path of odd length. This is because for any node, it must be able to have been colored in a path with alternating colors.

2 Thursday April 23rd

2.1 Scrap

2.1.1 Kruskal Algorithm Complexity

3 Friday April 24th

3.1 Puzzle Exercises

3.1.1 Lewis Carrol Puzzle

1. Yellow, Norwegian, Diplomat, Water, Fox
2. Blue, Italian, Doctor, Tea, Horse
3. Red, English, Sculptor, Milk, Snails
4. White, Spaniard, Violinist, Fruit Juice, Dog
5. Green, Japanese, Painter, Coffee, Zebra

3.1.2 Forty Ten Ten

Forty + Ten + Ten = Sixty

3.1.3 Send More Money

Send + More = Money

$$\text{Send} \tag{4}$$

$$+ \text{More} \tag{5}$$

$$= \text{Money} \tag{6}$$

$$D + E = Y + 10x_1 \tag{7}$$

$$x_1 + N + R = E + 10x_2 \tag{8}$$

$$x_2 + E + O = N + 10x_3 \tag{9}$$

$$x_3 + S + M = O + 10x_4 \tag{10}$$

$$x_4 = M \tag{11}$$

$$M = 1 \tag{12}$$

$$x_3 + S = O + 9 \tag{13}$$

Here, we split off between the possibility of $S = 8, 9$

First Scenario: $S = 8$

$$x_3 = O + 1 \tag{14}$$

$$O = 0, x_3 = 1 \tag{15}$$

$$x_2 + E = N + 10 \tag{16}$$

This is impossible because in this scenario, 0, 1 are both taken and therefore we can not get to 12

Second Scenario: $S = 9$

$$x_3 = O \tag{17}$$

$$x_3 = 0, O = 0 \tag{18}$$

$$x_2 + E = N \tag{19}$$

$$N = E + 1 \tag{20}$$

$x_2 = 1$ because if it is zero then $E = N$, which we cannot have

$$x_1 + E + 1 + R = E + 10 \tag{21}$$

$$x_1 + R = 9 \tag{22}$$

Since x_1 is equal to 0 or 1, R is either 8 or 9, which it isn't 9 since S has already taken it. $R = 8, x_1 = 1$.

Here we will need to split off possible values of E .

$$x_4 = 1, x_3 = 0, x_2 = 1, x_1 = 1 \quad (23)$$

$$M = 1, O = 0, R = 8 \quad (24)$$

First Scenario: $E = 2, N = 3$

$$E = 2, N = 3 \quad (25)$$

$$D + 2 = Y + 10 \quad D = Y + 8 \quad (26)$$

That can't be right, since 0 and 1 are taken so Y would have to be greater than 9 which it can not be.

Second Scenario: $E = 3, N = 4$

$$D + 3 = Y + 10 \quad (27)$$

$$D = Y + 7 \quad (28)$$

Impossible parce que $D = 9, Y = 2$

Third Scenario: $E = 4, N = 5$

$$D + 4 = Y + 10 \quad (29)$$

$$D = Y + 6 \quad (30)$$

C'est impossible parce que la seule manière pour laquelle ce equation est vrai c'est que D ou Y prennent chiffres qui sont déjà prisé.

Fourth Scenario: $E = 5, N = 6$

Fifth Scenario: $E = 6, N = 7$

3.1.4 Kernel

K is a kernel if and only if: every vertex in K has its successors outside of K and every vertex outside K has at least one of its successors in K .

Show that the algorithm (KERNEL algorithm) ends and that all vertices are colored. Each vertex has a number and is not part of a cycle. We visit each node from $i = 1, N$ and therefore color each all nodes along the way.

Prove the uniqueness of the kernel.

4 Wednesday April 29th

4.1 Exercises

4.1.1 Djiskra

5 Tuesday May 5th

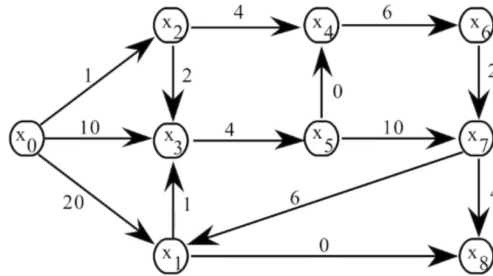
5.1 Maximum Flow

5.1.1 Exercise 1.1

Let N be the maximum number of cars per hour which can circulate this network.

Calculate N: the number of maximum number of cars per hour is 41.

We wish to find the values of the minimal paths from x_0 .



Apply DIJKSTRA algorithm. Write down the successive values of λ_i (if DIJKSTRA may be used), as well as a tree of minimal paths from x_0 .

Are the minimal paths unique? Justify your answer.

Figure 1

Step	x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
0 (start)	0	∞	∞	∞	∞	∞	∞	∞	∞
1 (x_0)	0	20	1	10	∞	∞	∞	∞	∞
2 (x_2)	0	20	1	3	5	∞	∞	∞	∞
3 (x_3)	0	20	1	3	5	7	∞	∞	∞
4 (x_4)	0	20	1	3	5	7	11	∞	∞
5 (x_5)	0	20	1	3	5	7	11	17	∞
6 (x_6)	0	20	1	3	5	7	11	13	∞
7 (x_7)	0	19	1	3	5	7	11	13	17
8 (x_8)	0	19	1	3	5	7	11	13	17
9 (x_1)	0	19	1	3	5	7	11	13	17

Table 1: Dijkstra's Algorithm

6 Wednesday May 6th

6.1 Linear Programming

6.1.1 Production Example

$$\text{maximize } \mathbf{f}^\top \mathbf{x} \quad (31)$$

$$\text{subject to } \mathbf{Ax} \preceq \mathbf{b} \quad (32)$$

$$\mathbf{x} \succeq \mathbf{0} \quad (33)$$

$$\text{maximize } f \quad (34)$$

$$\text{subject to } f = \sum_{j \in U^+(S)} x_{sj} - \sum_{j \in U^-(S)} x_{js} \quad (35)$$

$$(36)$$

7 Friday May 8th

7.1 Linear Programming

7.1.1 Exercise 7

A computer manufacturer must plan production over n months

- d_i = demand in month i
- r = regular-time capacity per month
-

8 Studying for Final Exam

8.1 2025 Exam Working

8.1.1 Exercise 1

The following graph G1 [2](#) is given. (Note: The nodes are labeled a, b, c, d and Fig. 1 represents Graph G1).

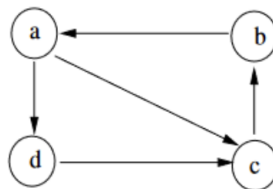


Figure 2

Provide the adjacency matrix and the alpha and beta tables. Find, if possible, a Hamiltonian circuit in the following graph. Justify your answer. Provide a coloring with the minimum number

of colors. Does a "good numbering" (topological sort) exist for the following graph (i.e., for every edge (i,j), number(i) < number(j))? If not, explain why it is not possible.

Solution: Adjacency matrix:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (37)$$

$$\beta = [c, d, a, b, c] \quad (38)$$

$$\alpha = [1, 3, 4, 5, 6] \quad (39)$$

A Hamiltonian circuit occurs starting at any of the nodes following the path formed by the following list of edges [a-d, d-c, c-b, b-a].

Beginning with a , we assign the color 1 to it and move onto b . b has a color of 2 due to it neighboring a , and subsequently we move onto c . c neighbors both $[a, b]$ and therefore has a color of 3. Finally, we visit d and assign a color of 2, since its neighbors are $[a, c]$. Three colors. We can create a fully connected subgraph of nodes $[a, b, c]$ and the paths that connect them. There are three nodes and therefore three colors is the upper and lower bound of the minimum number of colors to the graph. A good numbering does not exist because the graph has a cycle $[a, b, c, d]$.

8.1.2 Exercise 2

For the following graph 3: (Graph nodes: A, B, C, D, E with directed edge weights: B to A is 10, C to B is 20, A to C is 12, E to A is 7, A to D is 60, C to D is 32).

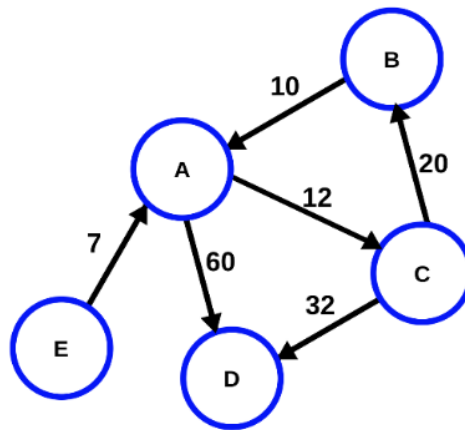


Figure 3

Find the shortest path originating from node E using Dijkstra's algorithm. Provide the Dijkstra table and the shortest path tree. Is this tree unique?

Solution: See solution in Table 2

The shortest path tree removes the arcs from B to A and that of A to D. This tree is indeed unique.

Step	E	A	B	C	D
0 (start)	0	∞	∞	∞	∞
1 (E)	0	7	∞	∞	∞
2 (A)	0	7	∞	19	67
3 (C)	0	7	39	19	51
4 (B)	0	7	39	19	51
5 (D)	0	7	39	19	51

Table 2: Dijkstra's Algorithm

8.1.3 Exercise 3

A farmer has a limited arable land area available and wishes to organize the production of two crops (e.g., tomatoes and peppers) in order to maximize his profit. Each crop requires a certain number of working hours and a certain amount of water, both available in limited quantities for the season. Furthermore, quotas may limit the area planted with one of the crops to respond to market conditions. The data is as follows:

He has a total of 20 hectares of land, 180 working hours, and 32 cubic meters of water. 1 hectare of tomatoes requires 5 working hours, 4 cubic meters of water, and yields a profit of 4 monetary units. 1 hectare of peppers requires 8 working hours, 2 cubic meters of water, and yields a profit of 6 monetary units. To stabilize the market, he cannot plant more than 6 hectares of tomatoes. Provide the linear programming formulation for the above problem. Propose the decision variables, the objective function, and the constraints.

Solution: For this problem, the quantity we seek to maximize is p , the total profit of the problem. This profit comes from the sales of tomatoes and peppers, of which the unit profit is provided for each. We define decision variables x_1, x_2 as the number of tomato and pepper hectares produced respectively. We seek to maximize the following:

$$p = 4x_1 + 6x_2 \quad (40)$$

There are also a few constraints associated with land, labor, water, non-negativity, and an additional tomato constraint. We get the following LP:

$$\text{maximize } 4x_1 + 6x_2 \quad (41)$$

$$\text{subject to } x_1 + x_2 \leq 20 \quad (42)$$

$$5x_1 + 8x_2 \leq 180 \quad (43)$$

$$4x_1 + 2x_2 \leq 32 \quad (44)$$

$$x_1 \leq 6 \quad (45)$$

$$x_1, x_2 \geq 0 \quad (46)$$

9 Thursday May 14th

9.1 Industrial Automation Exercise

9.1.1 Introduction

In this assignment, we address a **Three-Machine Flowshop Scheduling Problem** ($F3||\sum U_j$). The primary objective is to determine an optimal or near-optimal sequence of jobs to minimize the **number of tardy jobs**.

In a flowshop environment, every job J_j must be processed on Machine 1 (M_1), then Machine 2 (M_2), and finally Machine 3 (M_3) in that specific order. A job is considered tardy if its completion time on the final machine ($C_{j,3}$) exceeds its specified due date (d_j).

9.1.2 Problem Definition

We are given a set of 8 jobs, each with specific processing times $p_{i,j}$ (where i is the machine index and j is the job index) and a due date d_j .

9.1.3 Job Data

The following table provides the processing times for each machine and the respective due dates for 8 jobs:

Table 3: Processing Times and Due Dates for 8 Jobs

Job (j)	M_1 ($p_{1,j}$)	M_2 ($p_{2,j}$)	M_3 ($p_{3,j}$)	Due Date (d_j)
1	5	3	4	25
2	8	2	5	18
3	6	4	7	40
4	3	1	2	12
5	10	5	8	55
6	4	3	6	30
7	7	4	4	22
8	2	2	3	15

9.1.4 Mathematical Formulation

The objective is to minimize the total number of tardy jobs:

$$\min Z = \sum_{j=1}^8 U_j \quad (47)$$

Where:

$$U_j = \begin{cases} 1 & \text{if } C_{j,3} > d_j \\ 0 & \text{otherwise} \end{cases} \quad (48)$$

The completion time $C_{j,i}$ for job j on machine i is calculated as:

- $C_{j,1} = C_{j-1,1} + p_{1,j}$

- $C_{j,2} = \max(C_{j,1}, C_{j-1,2}) + p_{2,j}$
- $C_{j,3} = \max(C_{j,2}, C_{j-1,3}) + p_{3,j}$

Solution:

10 Challenge II

10.1 Work

Parameters: Below are the relevant parameters of the problem.

- **Site:** N is the number of production sites ($N = 30$). n is the index of production site, i.e. $n = 0$ refers to the first production site and $n = 29$ is the 30th.
- **Reseller:** M is the number of resellers, indexed by m from 0 to 3.
- **Time:** T is the time period, indexed by t , from 0 to 29.
- **Ingredient:** S is the number of ingredients, indexed by s , from 0 to 2, with 0 referring to sugar, 1 referring to powder milk, and 2 referring to flavour
- **Demand:** Each n production site has a daily demand for each product. This demand is $d_{n,t,s}$ referring to site, time, ingredient. Ex: $d_{20,3,2} = 20$ means the 21st site, in the fourth time period, has 20 units of demand for flavor.
- **Capacity:** Each reseller m has a capacity for each product s for each time period t $p_{m,s,t}$.
- **Inventory Cost:** Each n site has the ability to hold inventory at the end of a period t . We have an inventory cost $q_{n,s}$ as a parameter for the inventory holding cost associated with each site and ingredient.
- **Fixed Ordering Cost:** Each order has a fixed ordering cost parameter b_m that depends only on the reseller purchased from.
- **Variable Ordering Cost:** Each unit purchased as a cost c_s for each ingredient in the shipment, regardless of the seller.
- **Initial inventory:** We define an initial inventory for each site n and product s $v_{n,s}^0$

Variables: We can therefore create a set of variables to help us solve this problem.

- **Ordering Variable:** At each time step $t = 0, \dots, 29$, we must decide for each site $n = 0, \dots, 29$ how much of ingredient $s = 0, \dots, 2$ to order from each reseller $m = 0, \dots, 3$. We therefore create a variable $x_{n,m,t,s}$.
- **Inventory Stock:** We must also create a variable to denote, at the *beginning* of each time period t , how much inventory for each m ingredient to stock at each n site. This variable is $v_{n,t,s}$.
- **Fixed Order Binary:** $y_{n,m,t}$ to denote when we order from a manufacturer m to a site n in period t .

Objective: We want to minimize the total cost incurred by the company over the 30-day period while ensuring that each production site meets demand. This cost can be broken up into a few components.

$$\text{Inventory} + \text{Fixed Order Cost} + \text{Variable Cost} \tag{49}$$

Our variable cost will be constant, as we must meet demand over the course of each period, so we can exclude that from our problem. Inventory cost is the following equation:

$$\sum_N \sum_S q_{n,s} \sum_T v_{n,t,s} \quad (50)$$

In order to find the fixed order cost, we need to identify the number of orders we complete for each site n and seller m at each period t . We must create a binary variable $y_{n,m,t}$ to denote when we order from a manufacturer m to a site n in period t .

$$y_{n,m,t} = \begin{cases} 1 & \text{if } \exists s \in S \text{ such that } x_{n,m,t,s} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (51)$$

The fixed order cost is then

$$\sum_M b_m \sum_N \sum_T y_{n,m,t} \quad (52)$$

Our final objective is therefore

$$\text{minimize } \sum_N \sum_S q_{n,s} \sum_T v_{n,t,s} + \sum_M b_m \sum_N \sum_T y_{n,m,t} \quad (53)$$

Constraints:

$$\text{(Inventory balance)} \quad (54)$$

$$v_{n,t+1,s} = v_{n,t,s} + \sum_{m \in M} x_{n,m,t,s} - d_{n,t,s} \quad \forall n, t, s \quad (55)$$

$$\text{(Reseller capacity)} \quad (56)$$

$$\sum_{n \in N} x_{n,m,t,s} \leq p_{m,s,t} \quad \forall m, t, s \quad (57)$$

$$\text{(Fixed cost linking)} \quad (58)$$

$$\sum_{s \in S} x_{n,m,t,s} \leq M_{n,m,t} y_{n,m,t} \quad \forall n, m, t \quad (59)$$

$$\text{(Binary variables)} \quad (60)$$

$$y_{n,m,t} \in \{0, 1\} \quad \forall n, m, t \quad (61)$$

$$\text{(Non-negativity)} \quad (62)$$

$$x_{n,m,t,s} \geq 0, \quad v_{n,t,s} \geq 0 \quad \forall n, m, t, s \quad (63)$$

10.2 Final Solution

$$\text{minimize} \quad \sum_{n \in N} \sum_{s \in S} \sum_{t \in T} q_{n,s} v_{n,t,s} + \sum_{m \in M} b_m \sum_{n \in N} \sum_{t \in T} y_{n,m,t} \quad (64)$$

$$\text{subject to} \quad v_{n,t+1,s} = v_{n,t,s} + \sum_{m \in M} x_{n,m,t,s} - d_{n,t,s} \quad \forall n \in N, t \in T, s \in S \quad (65)$$

$$v_{n,0,s} = v_{n,s}^0 \quad \forall n \in N, s \in S \quad (66)$$

$$\sum_{n \in N} x_{n,m,t,s} \leq p_{m,s,t} \quad \forall m \in M, t \in T, s \in S \quad (67)$$

$$\sum_{s \in S} x_{n,m,t,s} \leq M_{m,t} y_{n,m,t} \quad \forall n \in N, m \in M, t \in T \quad (68)$$

$$x_{n,m,t,s} \geq 0, v_{n,t,s} \geq 0 \quad \forall n, m, t, s \quad (69)$$

$$y_{n,m,t} \in \{0, 1\} \quad \forall n \in N, m \in M, t \in T \quad (70)$$

11 Wednesday September 16th

We will have an exam on September 23rd at 8:00AM for linear programming. The final exam will be before the break and will be for the non-linear component of the course.

In the simplex algorithm, we need to start with an initial feasible basis. How do we get it? We have a two-step simplex algorithm, where the first step is to get a basic feasible solution of any sort, and the second step is to traverse from that basic feasible solution to get the global optima. In the first step, we add an artificial variable of the same size as the actual variables ($\mathbf{y}, \mathbf{x} \in \mathbb{R}^m$).

$$\text{minimize} \quad \mathbf{1}^\top \mathbf{y} \quad (71)$$

$$\text{subject to} \quad \mathbf{A}\mathbf{x} + \mathbf{y} = \mathbf{b} \quad (72)$$

$$\mathbf{x} \succeq 0, \mathbf{y} \succeq 0 \quad (73)$$

If we are able to get a solution where $\mathbf{1}^\top \mathbf{y} = 0$, then we have a basic feasible solution to the original problem. We can always get a basic feasible solution in this first step problem by setting $\mathbf{y} = \mathbf{b}$. After solving, we can remove the $\mathbf{y}$ variable, since it would be equal to zero anyways.

For the reduced cost, we get the cost in the context of the non-basic variables. We transform and then express the objective function by the non-basic variables.

11.0.1 Example 1

	x_1	x_2	x_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{2}$
x_1	1	$\frac{5}{4}$	0	$\frac{1}{2}$
$c^\top$	-4	-1	-1	0

	x_1	x_2	x_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{2}$
x_1	1	$\frac{5}{4}$	0	$\frac{1}{2}$
$c^\top$	0	$\frac{13}{4}$	0	$\frac{7}{2}$

	x_1	x_2	x_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{2}$
x_1	1	$\frac{5}{4}$	0	$\frac{1}{2}$
c^T	0	$\frac{13}{4}$	0	$\frac{7}{2}$

11.0.2 Example 2

$$\text{maximize} \quad -2x_2 + x_3 - x_4 \quad (74)$$

$$\text{subject to} \quad -x_2 + x_4 = 1 \quad (75)$$

$$\quad \quad \quad -x_3 + x_4 = 3 \quad (76)$$

$$\mathbf{x} \succeq 0 \quad (77)$$

To get a basic feasible solution we can add the variables y_1, y_2

$$\text{maximize} \quad -y_1 - y_2 \quad (78)$$

$$\text{subject to} \quad -x_2 + x_4 + y_1 = 1 \quad (79)$$

$$\quad \quad \quad -x_3 + x_4 + y_2 = 3 \quad (80)$$

$$\mathbf{x} \succeq 0, \mathbf{y} \succeq 0 \quad (81)$$

Basis	c_B	x_2	x_3	x_4	y_1	y_2	b
y_1	-1	-1	0	1	1	0	1
y_2	-1	0	-1	1	0	1	3
	c_j	0	0	0	-1	-1	
	z_j	1	1	-2	-1	-1	-4
	$c_j - z_j$	-1	-1	2	0	0	

We can pick x_4 to pivot on, and we can have y_1 leave the basis.

Basis	c_B	x_2	x_3	x_4	y_1	y_2	b
x_4	0	-1	0	1	1	0	1
y_2	-1	1	-1	0	-1	1	2
	c_j	0	0	0	-1	-1	
	z_j	-1	1	0	1	-1	-4
	$c_j - z_j$	1	-1	0	-2	0	

I will now stop this to move on to the next example since we moved on.

11.0.3 Example 3

$$\text{minimize} \quad x_1 + 6x_2 - 7x_3 + x_4 + 5x_5 \quad (82)$$

$$\text{subject to} \quad 5x_1 - 4x_2 + 13x_3 - 2x_4 + x_5 = 20 \quad (83)$$

$$\quad \quad \quad x_1 - x_2 + 5x_3 - x_4 + x_5 = 8 \quad (84)$$

$$\mathbf{x} \succeq 0 \quad (85)$$

We now want to test the feasibility so we add two variables y_1, y_2 and then solve simplex to minimize those artificial variables.

$$\text{maximize} \quad -y_1 - y_2 \quad (86)$$

$$\text{subject to} \quad 5x_1 - 4x_2 + 13x_3 - 2x_4 + x_5 + y_1 = 20 \quad (87)$$

$$x_1 - x_2 + 5x_3 - x_4 + x_5 + y_2 = 8 \quad (88)$$

$$\mathbf{x} \succeq 0, \mathbf{y} \succeq 0 \quad (89)$$

Basis	c_B	x_1	x_2	x_3	x_4	x_5	y_1	y_2	b
y_1	-1	5	-4	13	-2	1	1	0	20
y_2	-1	1	-1	5	-1	1	0	1	8
	c_j	0	0	0	0	0	-1	-1	
	z_j	-6	5	-18	3	-2	-1	-1	-28
	$c_j - z_j$	6	-5	18	-3	2	0	0	28

We choose x_1 to enter the basis and y_1 to leave

Basis	c_B	x_1	x_2	x_3	x_4	x_5	y_1	y_2	b
x_1	0	1	-4/5	13/5	-2/5	1/5	1/5	0	4
y_2	-1	0	-1/5	-8/5	-3/5	4/5	-1/5	1	4
	c_j	0	0	0	0	0	-1	-1	
	z_j	0	1/5	8/5	3/5	-4/5	1/5	-1	-4
	$c_j - z_j$	0	-1/5	-8/5	-3/5	4/5	-6/5	0	4

We choose x_5 to enter and y_2 to leave.

Basis	c_B	x_1	x_2	x_3	x_4	x_5	y_1	y_2	b
x_1	0	1	-4/5 + 1/20	13/5 + 3/5	-2/5 + 3/20	1/5 - 1/5	1/5 + 1/20	0 - 1/4	4 - 5/4
x_5	0	0	-1/4	-2	-3/4	1	-1/4	5/4	25/4
	c_j	0	0	0	0	0	-1	-1	
	z_j	0	1/5	8/5	3/5	-4/5	1/5	-1	-4
	$c_j - z_j$	0	-1/5	-8/5	-3/5	4/5	-6/5	0	4

Basis	c_B	x_1	x_2	x_3	x_4	x_5	y_1	y_2	b
x_1	0	1	-3/4	16/5	-1/4	0	1/4	-1/4	11/4
x_5	0	0	-1/4	-2	-3/4	1	-1/4	5/4	25/4
	c_j	0	0	0	0	0	-1	-1	
	z_j	0	1/5	8/5	3/5	-4/5	1/5	-1	-4
	$c_j - z_j$	0	-1/5	-8/5	-3/5	4/5	-6/5	0	4

11.0.4 Exercise 3.6

Find the optimal solution for the following LP:

$$\text{maximize} \quad 3x_1 + 3x_2 \quad (90)$$

$$\text{subject to} \quad x_1 + 4x_2 \leq 12 \quad (91)$$

$$2x_1 + x_2 \leq 10 \quad (92)$$

$$\mathbf{x} \succeq 0 \quad (93)$$

Solution: We can create our slack variables y_1, y_2 which also serve as our augmented variables for the phase 1 of the simplex since we are in $\mathbf{Ax} \succeq \mathbf{b}$ format.

I will jump into the tableau with the slack variables as the basis

Basis	c_b	x_1	x_2	y_1	y_2	b
y_1	0	1	4	1	0	12
y_2	0	2	1	0	1	10
	c_j	3	3	0	0	
	z_j	0	0	0	0	0
	$c_j - z_j$	3	3	0	0	0

here we can choose x_1 to enter the basis and y_2 to leave it.

Basis	c_b	x_1	x_2	y_1	y_2	b
y_1	0	0	7/2	1	-1/2	7
x_1	3	1	1/2	0	1/2	5
	c_j	3	3	0	0	
	z_j	3	3/2	0	3/2	15
	$c_j - z_j$	0	3/2	0	-3/2	-15

We choose x_2 to enter the basis and y_1 to leave it

Basis	c_b	x_1	x_2	y_1	y_2	b
x_2	3	0	1	2/7	-7	2
x_1	3	1	0	-1/7	-3	4
	c_j	3	3	0	0	
	z_j	3	3	3/7	-30	18
	$c_j - z_j$	0	0	-	-	-18

11.0.5 Exercise 3.8

Find the optimal solution for the following LP:

$$\text{maximize} \quad -3x_1 + 4x_2 \quad (94)$$

$$\text{subject to} \quad x_1 + 2x_2 \leq 18 \quad (95)$$

$$x_1 + x_2 \leq 10 \quad (96)$$

$$3x_1 + x_2 \leq 20 \quad (97)$$

$$x_1, x_2 \geq 0 \quad (98)$$

Solution:

Basis	c_b	x_1	x_2	x_3	x_4	x_5	b
x_3	0	1	2	1	0	0	18
x_4	0	1	1	0	1	0	10
x_5	0	3	1	0	0	1	20
	c_j	3	4	0	0	0	
	z_j	0	0	0	0	0	0
	$c_j - z_j$	3	4	0	0	0	0

We choose x_2 to enter the basis and x_3 to leave

Basis	c_b	x_1	x_2	x_3	x_4	x_5	b
x_2	4	1/2	1	1/2	0	0	9
x_4	0	1/2	0	-1/2	1	0	1
x_5	0	5/2	0	-1/2	0	1	11
	c_j	3	4	0	0	0	
	z_j	2	4	2	0	0	36
	$c_j - z_j$	1	0	-2	0	0	-36

We choose x_1 to enter the basis and x_4 to leave.

Basis	c_b	x_1	x_2	x_3	x_4	x_5	b
x_2	4	0	1	1	-1	0	8
x_1	3	1	0	-1	2	0	2
x_5	0	0	0	2	-5	1	6
	c_j	3	4	0	0	0	
	z_j	3	4	1	2	0	38
	$c_j - z_j$	0	0	-1	-2	0	-38

12 Wednesday September 23rd

12.1 Free Optimisation

Free optimisation corresponds to finding the optimal value of a function with variables that are unconstrained. We have $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = \frac{1}{2}ax^2 - bx$. We seek to find a $x^* \in \mathbb{R}$

$$\text{minimize } f(x) \quad (99)$$

To find the critical points of this function, we can take the derivative of the function

$$f(x) = \frac{1}{2}ax^2 - bx \quad (100)$$

$$\frac{df}{dx} = ax - b \quad (101)$$

When the derivative of a function is equal to zero, we get the critical points

$$x = \frac{b}{a} \quad (102)$$

We can also generalize to the multi-dimensional input space with a vector $\mathbf{x} \in \mathbb{R}^n$

$$\text{minimize } f(x) = \frac{1}{2}\mathbf{x}^\top \mathbf{A}\mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad (103)$$

This is the quadratic form of f . If the gradient of this quadratic form is equal to zeros, then we are at a critical point

$$\nabla f(x) = \mathbf{A}\mathbf{x} - \mathbf{b} = \mathbf{0} \quad (104)$$

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{h}) - f(\mathbf{x})}{t} = \nabla f(x^*)^\top \mathbf{h} \quad (105)$$

12.1.1 General Case

We can have the general case $f : \mathbb{R}^n \rightarrow \mathbb{R}$ where our only constraint is that f is differentiable. We want to find the optimal solution $\mathbf{x}^*$ that minimizes $f(\mathbf{x}^*)$. The solution must occur where the derivative of the function is zero.

12.2 Exercises

12.2.1 First

We have $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. With the form $f(x, y) = 8x^2 + y^2 + xy + 1$.

Solution: We can solve this by taking the partial derivatives and stacking them or doing the limit with $\mathbf{l} = [h, k], t$

$$\frac{f((x, y) + t(h, k)) - f(x, y)}{t} = \frac{f(x + th, y + tk) - f(x, y)}{t} \quad (106)$$

$$= 8(k + th)^2 + (y + tk)^2 + (x + th)(y + tk) + 1 - (8x^2 + y^2 + xy + 1) \quad (107)$$

$$\dots \quad (108)$$

$$\nabla f(x, y) = \begin{bmatrix} 16x + y \\ 2y + x \end{bmatrix} \quad (109)$$

The partial derivative method is much simpler.

12.2.2 Second

$$f(x, y) = x^2 + y^2 + \frac{1}{x^2 y^2}$$

$$\frac{df}{dx} = 2x - \frac{2}{x^3 y^2} \quad (110)$$

$$\frac{df}{dy} = 2y - \frac{2}{x^2 y^3} \quad (111)$$

$$\nabla f(x, y) = \begin{bmatrix} 2x - \frac{2}{x^3 y^2} \\ 2y - \frac{2}{x^2 y^3} \end{bmatrix} \quad (112)$$

12.2.3 Third

$$f(x, y) = (2x - x^2)(2y - y^2)$$

$$\frac{df}{dx} = (2 - 2x)(2y - y^2) \quad (113)$$

$$\frac{df}{dy} = (2 - 2y)(2x - x^2) \quad (114)$$

$$\nabla f = \begin{bmatrix} (2 - 2x)(2y - y^2) \\ (2 - 2y)(2x - x^2) \end{bmatrix} \quad (115)$$

The solutions occur at $x = 0, y = 0, x = 1, y = 1, x = 2, y = 2$

12.2.4 Fourth

$$f(x, y, z) = 8x^2 + y^2 + z^2 + xz + yz + 1$$

$$\frac{df}{dx} = 16x + z \quad (116)$$

$$\frac{df}{dy} = 2y + z \quad (117)$$

$$\frac{df}{dz} = 2z + x + y \quad (118)$$

$$\nabla f = \begin{bmatrix} 16x + z \\ 2y + z \\ 2z + x + y \end{bmatrix} \quad (119)$$

We can get the solution here by solving the linear equations $\mathbf{Ax} = \mathbf{0}$ where

$$\mathbf{A} = \begin{bmatrix} 16 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad (120)$$

12.2.5 fifth

$$f(x, y, z, t) = t(x^2 + y^2) + z^3$$

$$\frac{df}{dx} = 2tx \quad (121)$$

$$\frac{df}{dy} = 2ty \quad (122)$$

$$\frac{df}{dz} = 3z^2 \quad (123)$$

$$\frac{df}{dt} = x^2 + y^2 \quad (124)$$